

Backend boshlanishi: Network, TCP va HTTP bo'yicha uyga vazifa

Masala 1. Go dasturida TCP server yozing. Server 8080 portda tinglasin va client ulanadigan bo'lsa, unga "Salom, TCP client!" matnini yuborsin.

Masala 2. Go dasturida TCP client yozing. Client 127.0.0.1:8080 manziliga ulansin, serverdan kelgan xabarni o'qib terminalga chiqarsin.

Masala 3. TCP serverni shunday o'zgartiring: client yuborgan matnni o'qisin va aynan shu matnni "echo: <matn>" ko'rinishida qaytarsin.

Masala 4. TCP client terminaldan foydalanuvchi kiritgan bitta satrni serverga yuborsin va server javobini terminalga chiqarsin.

Masala 5. TCP server bir nechta client bilan ishlay olishi uchun har bir ulanishni alohida goroutine ichida qayta ishleng.

Masala 6. TCP serverda client bo'sh satr yuborsa, ulanishni yopadigan mantiq qo'shing. Client tomonida ham server ulanishni yopganini chiroyli xabar bilan ko'rsating.

Masala 7. TCP server har bir client xabariga hozirgi vaqtni qo'shib javob bersin. Masalan: "2026-08-25 14:30:00 | echo: salom".

Masala 8. TCP orqali oddiy chatga o'xshash dastur yozing: client bir nechta xabar yubora olsin, server esa har biriga javob qaytarsin. Client "exit" yozsa dastur tugasin.

Masala 9. TCP serverga timeout qo'shing: agar client 10 soniya ichida hech qanday xabar yubormasa, server ulanishni yopib yuborsin.

Masala 10. TCP server clientdan kelgan xabar uzunligini tekshirsin. Agar xabar 100 belgidan uzun bo'lsa, "xato: xabar juda uzun" deb javob qaytarsin.

Masala 11. Go standart kutubxonasi yordamida HTTP server yozing. Server /hello endpointiga GET so'rov kelganda "Hello, HTTP!" javobini qaytarsin.

Masala 12. HTTP serverda /time endpointini yarating. Bu endpoint hozirgi vaqtni plain text ko'rinishida qaytarsin.

Masala 13. HTTP serverda /user endpointini yarating. Query parametrdan name qiymatini o'qib, "Salom, <name>" javobini qaytarsin. Agar name berilmasa, "Salom, mehmon" qaytsin.

Masala 14. HTTP serverda faqat POST metodini qabul qiladigan /echo endpointini yarating. Request body ichidagi matnni o'qib, javob sifatida qaytarsin.

Masala 15. HTTP serverda noto'g'ri metod ishlatilsa 405 Method Not Allowed statusini qaytaring. Masalan, /echo endpointiga GET yuborilsa 405 qaytsin.

Masala 16. HTTP server JSON javob qaytaradigan /status endpointini yarating. Javob: {"service":"lesson20","status":"ok"}. Content-Type application/json bo'lsin.

Masala 17. HTTP client yozing. Client /status endpointiga GET so'rov yuborsin, status code va response body ni terminalga chiqarsin.

Masala 18. HTTP serverda /headers endpointini yarating. U request headerlaridan User-Agent qiymatini o'qib, javobda qaytarsin.

Masala 19. HTTP serverga oddiy middleware qo'shing. Har bir request uchun method, path va request bajarilgan vaqtni terminalga log qiling.

Masala 20. Mini loyiha: TCP va HTTP bo'yicha o'rganganlaringiz asosida kichik "note service" yozing. HTTP orqali /notes endpointiga POST bilan note qo'shilsin, GET bilan barcha notelar JSON ko'rinishida qaytsin. Notelar xotirada slice ichida saqlansin.